

PROFILE

Passionate second-year PhD student in Computer Science and Engineering looking for intern software/system engineering positions with experiences in sparse tensor algebra accelerations, GPU code development, and knowledge in AI/ML.

EDUCATION

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|-----------------------------------------------------------------|---------------------|
| Ph.D. Computer Science, University of California - Riverside    | Sep 2024 – Current  |
| Riverside, California                                           |                     |
| B.S. Computer Engineering, University of California - San Diego | Sep 2019 – Sep 2023 |
| San Diego, California                                           |                     |

PROJECTS & CONTRIBUTIONS

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| RTSpMSpM: Harnessing Ray Tracing for Efficient Sparse Matrix Computations                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | September 2023 – Present |
| Lab: ESCAL, Riverside                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                          |
| <ul style="list-style-type: none"><li>Designed the algorithm that maps the sparse matrix multiplications (SpMSpM) into ray tracing problems to enable the use of ray tracing accelerators in solving SpMSpM.</li><li>Developed a library that realized the proposed algorithm using OptiX and CUDA to achieve 1.85× speedup over cuSPARSE on NVIDIA’s RT Cores.</li><li>Integrated the developed library into a graph neural network training framework using CUDA and Python to achieve more than 3× speedup on NVIDIA RTX 4090 GPUs.</li></ul> |                          |
| Machine Learning in Music                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | June 2023 – August 2023  |
| <ul style="list-style-type: none"><li>Developed a system using PyTorch and autoencoder network to generates music clips based on the audio genre within 3 minutes using LSTM.</li><li>Built an audio genre classification system using CNN and spectrograms.</li><li>Trained autoencoder networks for feature compression and denoising.</li></ul>                                                                                                                                                                                               |                          |

TECHNICAL SKILLS

- Programming and Scripting Languages**  
Python, C++, C, Bash
- Libraries and Frameworks**  
NVIDIA OptiX, CUDA, cuSPARSE, NumPy, Docker
- Tools and Debuggers**  
GitHub, gdb, jdb, vim
- Operating Systems and Simulation Software**  
Linux (Ubuntu), Windows, MATLAB, LTSpice, Multisim, Proteus

PUBLICATIONS

**Zhang Hongrui, Zhang Yunan, Tseng Hung-Wei.** (2025). *RTSpMSpM: Harnessing Ray Tracing for Efficient Sparse Matrix Computations*. In *Proceedings of the 52nd Annual International Symposium on Computer Architecture (ISCA '25)*, pages 359-373. DOI: 10.1145/3695053.3731072